

1 Inverse Matrices

1. If square matrix A has an inverse, then, both $A^{-1}A = I$ and $AA^{-1} = I$.
2. The algorithm to test invertibility is elimination : A must have n pivots (nonzero).
3. The algebra test for invertibility is the determinant of A : the determinant of A must be not zero $\det(A) \neq 0$.
4. The equation that for invertibility is : $Ax = 0$, the x must only be the 0.
5. If A and B (same size) are invertible then, so is AB also invertible, $(AB)^{-1} = B^{-1}A^{-1}$
6. $A^{-1}A = I$ is n equations for n columns of A^{-1} , Gauss-Jordan eliminates $[AI]$ to $[IA^{-1}]$.

1.1 Inverse Matrix

Definition : the matrix A is invertible if there exists a matrix A^{-1} that “inverts” A :

$$A^{-1}A = I$$

and

$$AA^{-1} = I$$

Not all Matrices have inverse.

- Note 1 : The inverse exists if and only if elimination produces n pivots (row exchanges are allowed.)
Acutally, the elimination solves $Ax = b$ without explicitly use the Inverse A^{-1} .

$$A^{-1}Ax = A^{-1}b \Rightarrow x = A^{-1}b$$

- Note 2 : The matrix A only have one inverse : if $AC = I$ and $BA = I$ then ,

$$C = B = A^{-1}$$

- Note 3 : If A is invertible, the one and the only solution to $Ax = b$ is

$$x = A^{-1}b$$

- Note 4 : Suppose there is a nonzero vector x ($x \neq 0$) such that : $Ax = 0$. then, **the A is not invertible (haven't inverse)**. as no matrix can bring 0 back to x (nonzero). So, **If $Ax = 0$ then, the only solution must is $x = 0$.**
- Note 5 : A is a 2 by 2 matrix, and is invertible and only if the determinant $ad - bc$ of the matrix A is not zero, $\det(A) = ad - bc \neq 0$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- Note 6 : A digonal matrix has an inverse provied no digonal entries are zero :

If

$$A = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix}$$

then,

$$A = \begin{bmatrix} \frac{1}{d_1} & & \\ & \ddots & \\ & & \frac{1}{d_n} \end{bmatrix}$$

2 The inverse of a Product AB

For the inverse of a Product AB is come from **reverse order** :

If A and B are invertible then, so is AB , the inverse of a Product AB is

$$(AB)^{-1} = B^{-1}A^{-1}$$

> “If you put on socks and then shoes, the first to be taken off are the shoes. “

Inverse of AB :

$$(AB)(B^{-1}A^{-1}) = AIA^{-1} = AA^{-1} = I$$

Same as the three case :

$$(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$$

that because of :

$$(C^{-1}B^{-1}A^{-1})ABC = I$$

3 Calculating A^{-1} bt Gauss-Jordan Elimination

3.1 1. The Core Idea

Gilbert Strang emphasizes that matrix multiplication works by columns. If A multiplied by its inverse A^{-1} yields the identity matrix I , we are essentially solving n systems of equations simultaneously:

$$AA^{-1} = A[x_1 \ x_2 \ \dots \ x_n] = [e_1 \ e_2 \ \dots \ e_n] = I$$

Where:

- x_j is the j -th column of A^{-1} .
- e_j is the j -th column of the identity matrix (a vector with 1 in the j -th row and 0 elsewhere).

Instead of solving $Ax_j = e_j$ separately for each column, the Gauss-Jordan method solves them all at once by constructing a massive augmented matrix.

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3.2 2. The Augmented Matrix Setup

We multiply the size of our workspace by pinning A and I side-by-side:

$$[A \mid I] = \left(\begin{array}{cccc|cccc} a_{11} & a_{12} & \dots & a_{1n} & 1 & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & a_{2n} & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} & 0 & 0 & \dots & 1 \end{array} \right)$$

Using row operations (elimination), we clear out the entries below **and above** the pivots. Our goal is to reduce the left side (A) to the identity matrix I .

As Strang famously summarizes: Multiply by A^{-1} on the left to see what happens.

$$A^{-1}[A \mid I] = [I \mid A^{-1}]$$

If elimination succeeds in producing I on the left, the inverse automatically appears on the right.

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3.3 3. Step-by-Step Gauss-Jordan Algorithm

1. **Forward Elimination:** Produce zeros below the pivots to reach an upper triangular matrix U on the left half.
2. **Backward Elimination:** Work from the bottom row upward, using the pivots to clear out entries **above** them to reach a diagonal matrix D .
3. **Normalize:** Divide each row by its respective pivot to turn the diagonal elements into 1s, completing the transition from D to I .

> **Note on Invertibility:** If elimination encounters a zero in a pivot position that cannot be fixed by swapping rows, the matrix is singular (it has no inverse).

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3.4 4. Worked Example: A 2×2 Matrix

Let's find the inverse of $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$.

3.4.1 Step 1: Set up the augmented matrix $[A \mid I]$

$$\left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right)$$

3.4.2 Step 2: Clear below the first pivot

Subtract $\frac{1}{2}$ of Row 1 from Row 2 ($R_2 \leftarrow R_2 - \frac{1}{2}R_1$):

$$\left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 1 \end{array} \right)$$

3.4.3 Step 3: Clear above the second pivot (Upward Elimination)

Subtract 6 times Row 2 from Row 1 ($R_1 \leftarrow R_1 - 6R_2$):

$$\left(\begin{array}{cc|cc} 2 & 0 & 4 & -6 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 1 \end{array} \right)$$

3.4.4 Step 4: Divide by the pivots to get I on the left

Divide Row 1 by 2, and multiply Row 2 by 2:

$$\left(\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & 1 & -1 & 2 \end{array} \right)$$

Therefore, the inverse matrix is:

$$A^{-1} = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

4 Singular versus Invertible

- A^{-1} exists exactly when A has a full set of n pivots.

(Row exchanges are allowed.) Prove by Gauss-Jordan Elimination :

1. With n pivots, elimination solves all equations $Ax_i = e_i$, the columns x_i go into A^{-1} . Then $AA^{-1} = I$ and A^{-1} is the **right reverse**.
2. Elimination is really a sequence of multiplications by E and P and D^{-1} :

$$CA = (D^{-1} \dots E \dots P \dots E)A = I$$

D^{-1} divides by the pivots. The Matrices E produce zeros below and above the pivots . and the matrix P will exchange the rows (if needed). the product matrix in equation CA is evidently (显然地) a **left-inverse of A**

- Reasoning in reverse will show that A must have n pivots if $AC = I$:

1. If A don't have n pivots, elimination will lead to a zero row.
 2. Those elimination setps are taken by an invertible M . so a row of MA is zero.
 3. If $AC = I$ had been possible, then, $MAC = M$ the zero row of MA times the matrix C gives a zeros row of M ifself.
 4. An invertible matrix M can't have a zeros row, so the matrix A must have n pivots if $AC = I$.
- so the conclulusion is : If $AC = I$ then $CA = I$ and the $C = A^{-1}$.

- Note 1 : If A is a triangular matrix, if and only if no diagonal entries are zero.

5 Recognizing an Invertible Matrix

The matrix A is invertible when **Diagonally dominant matrices** : Each a_{ii} on the diagonal is larger than the total sum along the rest of row i . On every row , :

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

Example : A is Diagonally dominant ($3 > 2$) , B is not (but still invertible), C is singular .

$$A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

- Reasoning :

Take any nonzero vector x , suppose its largest component is $|x_i|$. Then $Ax = 0$ is impossible, because the row i of $Ax = 0$ would need :

$$a_{i1}x_1 + \dots + a_{ii}x_i + \dots + a_{in}x_n = 0$$

Those can't add to zero when A is Diagonally dominant. the size of $a_{ii}x_i$ is greater than all the other terms combined :

$$\begin{aligned} |x_j| &< |x_i| \\ \sum_{j \neq i} |a_{ij}x_j| &\leq \sum_{j \neq i} |a_{ij}| |x_i| < |a_{ii}| |x_i| \end{aligned}$$

This shows that the $Ax = 0$ is only possible when $x = 0$, so A is invertible